

Project Design Phase

Solution Architecture

Date	15-07-2026
Team ID	23BFA05428
Project Name	Smart Library Request Workflow in ServiceNow
Maximum Marks	4 Marks

Solution Architecture:

The Smart Library Request Workflow is built on the ServiceNow platform using a layered architecture. Students interact with the system through a Service Catalog item that captures the book request, while librarians use a dedicated management interface to maintain the catalogue and control the approval process. The application logic layer validates each request, checks role-based access and book availability, and routes it through an automated approval workflow before updating stock records and notifying the requester.

Layered Components:

- **User Interaction Layer:** The Service Catalog provides a self-service interface where students raise book requests and track status, while librarians access a management interface to review pending requests and maintain the catalogue.
- **Business Logic Layer:** Catalog Client Scripts, Flow Designer workflows, and Script Includes work together to validate requests, enforce role-based access control, verify book availability, and route requests for librarian approval or rejection.
- **Data & Service Layer:** ServiceNow tables store book, request, and inventory records, while the Notification Engine, inventory update logic, and Dashboards provide status alerts, stock synchronization, and analytics on borrowing trends.

Solution Architecture Diagram:

